

Axona Quick Start

Get a working pub/sub roundtrip in **under five minutes** on the current **@axona/protocol v4.11.2** API (kernel 4.11.2). One Node process connects to the live **testnet** bridge, subscribes to an open topic, publishes a signed message, and logs what comes back.

Testnet, for now. The 4.x line runs on **testnet** (`wss://testnet.axona.net`). The production bridges (`wss://bridge.axona.net`) are still on the 3.x line, and the two don't interoperate (the wire major partitions them). So this Quick Start uses testnet + the **#v4.11.2** pin. To target production instead, install a 3.x tag and point at `wss://bridge.axona.net`.

Companion documents:

- API Reference — every exported symbol.
- Programmer Guide — mental model + worked example.

Prerequisites

- **Node.js 20+** (for built-in Web Crypto Ed25519)
- A terminal with network access (we connect to `wss://testnet.axona.net`)

No build step, no DB, no local bridge — the testnet bridge is the entry point.

1. Install (30 seconds)

```
mkdir my-axona-demo && cd my-axona-demo
npm init -y
npm pkg set type=module
npm install github:axona-net/axona-protocol#v4.11.2
```

2. Two identities, one rule

Axona separates **who you connect as** from **who you publish as**:

- `createNodeIdentity({ lat, lng })` -> your **connection** identity. Its public key forms the 66-hex node ID (`.id`); the leading byte encodes your region. This authenticates the transport. It is fine to mint a fresh one each run.
- `createAuthorIdentity({ persistAs? })` -> your **author** identity. Its public key is the 64-hex Author ID (`.authorId`), which appears on the wire as **signerPubkey**. It has no location and no node ID — authorship is not a place. Pass `persistAs` to keep a stable author across runs.

The one rule: every publish names its signer via `{ signWith }` — an author identity, or the **ANONYMOUS** sentinel for a deliberately unsigned post. There is no default signer, and the node key never signs publishes.

3. Topics are descriptors, not strings

A topic is a small descriptor `{ region?, owner?, name, write? }`; its 66-hex topic ID is a hash of that descriptor. Because the ID is a pure function of the fields, **anyone who knows the fields computes the same ID** — no registry, no coordination.

- **region** — a real geographic cell. Accepts a name (`'useast'`), a hex string (`'0x89'`), or a number (137) — all equivalent. Omit it and it defaults to the publisher's own node region; name it to share a topic across regions.
- **owner** — an Author ID, or absent.
- **name** — the human label (`'lobby'`, `'profile'`).
- **write** — `'open'` or `'owner'`. **Defaults by owner:** no owner -> `'open'` (anyone publishes); an owner -> `'owner'` (only the owner key publishes, the safe default). With no owner, **write** is ignored — you can't have an owner-only topic with no owner.

This Quick Start uses an **open** topic { **region**, **name** }: anyone publishes, anyone subscribes. (Note: the removed v2 model — string topics, a **publisher** argument, **sign:false**, **geoCellId/regionSynthPublisher** synthetic publishers — is gone. Topics are descriptors now.)

`deriveTopicId(descriptor)` returns the 66-hex ID — the shareable read handle. **sub** accepts **either** a descriptor **or** that ID; **pub** always needs the descriptor (the storing node recomputes the ID to enforce the write policy).

4. Write the demo (one file)

Save this as `index.js` — it mirrors how `apps/axona-minimal` wires a peer:

```
import {
  AxonaPeer, AxonaDomain, NeuronNode,
  createNodeIdentity, createAuthorIdentity,
  deriveTopicId, KERNEL_VERSION,
} from '@axona/protocol';
import { webTransport } from '@axona/protocol/transport/web/index.js';

const BRIDGE = 'wss://testnet.axona.net';           // live testnet bridge (kernel 4.11.2)
const HERE   = { lat: 38.0, lng: -77.0 };           // your real location (us-east here)
const TOPIC  = { region: 'useast', name: 'quick-start-demo' }; // open topic

// 1. Two identities: connection (node) + authorship (author).
const node$identity = await createNodeIdentity(HERE); // ephemeral connection key
const author        = await createAuthorIdentity();   // ephemeral author key

// 2. Build the peer on the web transport pointed at the bridge.
const transport = webTransport({ bridgeUrl: BRIDGE, identity: node$identity });
const node      = new NeuronNode({ id: BigInt('0x' + node$identity.id), lat: HERE.lat, lng: HERE.lng });
node.transport = transport;
const peer = new AxonaPeer({
  domain: new AxonaDomain({ k: 20 }),
  node,
  nodeIdentity: node$identity,
  transport,
});

await transport.start(node$identity.id);
await peer.start();

// 3. Wait for the synaptome (routing mesh) to warm up before pub/sub.
console.log('kernel v' + KERNEL_VERSION + ' - connecting...');
const until = Date.now() + 30000;
while (Date.now() < until && (node.synaptome?.size ?? 0) < 3) {
  await new Promise((r) => setTimeout(r, 600));
}
console.log('mesh ready (' + (node.synaptome?.size ?? 0) + ' peers)');
console.log('topic id:', await deriveTopicId(TOPIC));

// 4. Subscribe. The handler gets an envelope: { msgId, seq, ts, topic, message, signerPubkey? }.
const sub = await peer.sub(TOPIC, (env) => {
  console.log(['recv'], env.message, 'from', (env.signerPubkey || 'anon').slice(0, 12));
}, { since: 'all' });
```

```
// 5. Publish, naming the signer. signWith is REQUIRED.
const msgId = await peer.pub(TOPIC, 'hello from the quick start', { signWith: author });
console.log('[pub ] msgId=' + msgId.slice(0, 12) + '...');

// 6. Give fan-out a moment, then leave cleanly.
await new Promise((r) => setTimeout(r, 2000));
await sub.stop();
await peer.leave();
process.exit(0);
```

5. Run

node index.js

You should see something like:

```
kernel v4.11.2 - connecting...
mesh ready (4 peers)
topic id: 89a1b2c3...
[pub ] msgId=8e9d4b1a30c2...
[recv] hello from the quick start from 4f2a9b0c1d3e
```

That's it. You just:

- minted an Ed25519 connection identity (66-hex node ID, region-prefixed) and a location-free author identity (64-hex Author ID = `signerPubkey`),
- connected to the live bridge over the web transport and warmed up a routing mesh,
- derived a deterministic topic ID from `{ region, name }`,
- published a **signed** envelope to the topic's root axons and received it back through a `since: 'all'` subscriber.

What just happened

<pre>peer.pub({region,name}, msg, {signWith: author}) v deriveTopicId({region, name}) ----- same fields -----> (identical 66-hex topic id on both sides) v build signed envelope -> route to topic's K-closest root axons (publish-k)</pre>	<pre>peer.sub({region,name}, handler, {since:'all'}) v deriveTopicId({region, name}) v subscribe-k to the same K-closest root axons v +----- fan-out -----> handler(env): { msgId, seq, ts, topic, messa</pre>
---	---

Both sides compute the identical topic ID because the ID is a pure hash of the descriptor fields. That ID-matching is the rule you can't break — same `{ region, name }`, same topic.

Going further

Want to...	Do this
Make authorship stable across runs	<code>createAuthorIdentity({ persistAs: 'me' })</code>
Publish anonymously	<code>peer.pub(topic, msg, { signWith: ANONYMOUS })</code> (import <code>ANONYMOUS</code>)
Own a feed only you can write	<code>{ region, owner: me.authorId, name: 'profile' }</code> (write defaults to <code>'owner'</code>)

Want to...	Do this
Share a read-only handle	<code>await deriveTopicId(descriptor)</code> -> hand out the 66-hex ID; <code>sub/pull/metrics</code> accept it
Run against production (3.x)	install a 3.x tag and set <code>BRIDGE = 'wss://bridge.axona.net'</code> (prod is a separate, non-interoperating line)
See the full mental model	Programmer Guide
Look up a specific symbol	API Reference

Troubleshooting

Cannot find module '@axona/protocol' — make sure `package.json` has `"type": "module"` and the install completed. Re-run `npm install`.

Cannot mix BigInt and other types — `NeuronNode` XORs node IDs as `BigInts`. Pass `BigInt('0x' + node$identity.id)`, not the raw hex string.

peer.pub: name a signer... — `signWith` is required. Pass an author identity, or `{ signWith: ANONYMOUS }` for an unsigned publish. There is no default signer.

Connects but nothing arrives — you almost certainly published before the mesh warmed up. Subscribing/publishing on a cold synaptome routes to too few root axons. Wait for `node.synaptome.size` to reach a few peers (the warm-up loop above) before calling `pub/sub`. On the public bridge this takes a few seconds.

Can't reach the bridge / connection hangs — confirm outbound `wss://` (TLS WebSocket) to `testnet.axona.net` is allowed by your network. There is no fallback to a local process; the demo needs the bridge to find peers.

UPGRADE_REQUIRED close code (4426) — a wire/version mismatch. The testnet bridge runs kernel 4.11.2 (wire 4.0); install `github:axona-net/axona-protocol#v4.11.2`. Note this also fires if you point a 4.x peer at a **production** (3.x) bridge — they're a hermetic wire partition, so match the pin to the network.